Signed FFN Layer-Order Context for Neural Spectral Capacity

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Research question

Zhu, Zhao and Lu's additive Neural Spectral Capacity (NSC) cannot distinguish FFN-width reorderings at fixed hidden width. Within Section 3's core scope, we test whether preceding FFN context improves ranking and budgeted selection, and whether increasing or decreasing weights work better.

Method and mechanism checks

Keep starter attention/FFN capacities A_l, F_l, matrix grouping and $s = 0.02$ fixed. With hidden/FFN widths d_l, f_l, define our specification-only heuristic:

$$q_l = ½\, d_l\, f_l\, s^4,\quad Q_l = \sum_{j<l} q_j,\quad w(Q;\beta) = 1 + \beta Q/(1 + |\beta| Q),$$

$$S(\beta) = \sum_l [A_l + w(Q_l;\beta)\, F_l].$$

For $Q \geq 0$, $\beta < 0$ recovers reciprocal damping; $\beta > 0$ gives increasing weights; $\beta = 0$ recovers NSC (ablation). Weights satisfy $0 < w < 2$ and $w(0;\beta) = 1$. The proxy q_l is motivated by the second-moment ratio of a bias-free, unnormalized ReLU branch with independent Gaussian weights and isotropic inputs. In this Post-LN archive, Q_l is structural context, not accumulated activation variance. Check unlabelled reorderings, zero and limiting cases; sensitivity alone does not establish prediction. Joint attention/FFN weighting and exponential damping test weighting scope and form.

Data and evaluation

Use 200 archived Transformers from Javaheripi et al.'s LiteTransformerSearch: 100 development and 100 final architectures. Select β from $\{0, \pm0.1, \pm1, \pm10, \pm100, \pm1000\}$ by development Spearman correlation with negative validation perplexity (PPL); break ties by smaller $|\beta|$, then negative sign. Report both signs' sensitivity. Freeze choices and code before final test-PPL evaluation. Baselines: NSC/ablation, positional example, reported total parameters, reference FLOPs and uniform random selection.

Evaluation	Protocol and safeguards
Ranking	Spearman ρ, Kendall τ_b against negative PPL, and ties; same-hidden-width pair comparisons with total-parameter ratio min/max ≥ 0.90 and pair counts. Bootstrap architectures, rebuilding overlapping pairs.
Budgeted selection	Total-parameter caps: 10M/20M/30M/50M; $k = 1, 3$, identical pools. Report pool sizes, best selected PPL and regret (selected minimum minus feasible minimum); 200 fixed-seed selection/tie-breaking trials.
Embedding effects	NSC excludes embeddings. Compare identical embedding settings; report group sizes, with descriptive summaries for small groups. Retain all main budgets.

Compute and deliverables

Laptop CPU, Python 3.9+; no new training, GPU or LLM API. Reproduce the starter, implement/check the score and ablation, then develop/freeze and evaluate. Deliver reproducible code, configurations, commands, commit/tag, scoring cost and a report. One outcome per architecture precludes training-seed uncertainty estimates; this heterogeneous panel cannot establish causal FFN-order effects. Negative results remain informative.